



Detecting Implicit Biases of LLMs with Bayesian Hypothesis Testing

GROUP

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● ChatGPT-3.5-Turbo ● DeepSeek-V3

Motivation & Research Question

LLMs (GPT, DeepSeek, Llama...) **inherit social biases and stereotypes** embedded in their training corpora. Prior work — biased word embeddings (Bolukbasi 2016), human-like associations (Caliskan 2017) — *measured* the effect but rarely *tested* it.

RESEARCH QUESTION

Can we **formally test** — not just measure — whether an LLM is biased for a given social category, with a method that can also quantify evidence *for the absence* of bias?

THE PAPER'S KEY IDEA

Reformulate bias detection as a **hypothesis test** on binary-choice questions.

One framework unifies the **frequentist p-value** and the **Bayesian Bayes factor**.

Applied to **ChatGPT-3.5** and **DeepSeek-V3** across three benchmark datasets.

Statistical Framework

everything is a function of (n, k)

STEP 1

Ask n binary-choice questions:
stereotypical vs anti-stereotypical
option



STEP 2

Record each answer as an indicator:
 $X = 1$ if stereotypical, else 0



STEP 3

Tally to (n, k)
 $k = \sum X_i = \#$ stereotypical picks

SAMPLING MODEL

$$k \sim \text{Binomial}(n, \pi)$$

each answer i.i.d. Bernoulli(π); π = model's true probability of the
stereotypical choice

HYPOTHESES

$$H_0: \pi = 0.5$$

no bias — fair coin

vs

$$H_1: \pi \neq 0.5$$

bias present

Three Metrics from (n, k)

implemented in `src/R/stats.R`

1 · Stereotype Score

$$SS = \max\{k/n, 1 - k/n\}$$

magnitude only, always ≥ 0.5 , **direction-agnostic**. Its signed companion is `rate = k/n`.

```
R stereotype_score(n,k)
```

```
R stereotype_rate(n,k)
```

2 · Exact Binomial Test

$$p = P(S_n \leq LB) + P(S_n \geq UB)$$

frequentist two-sided p for H_0 . Flags ***** $p < 0.05$, ****** $p < 0.01$. Can reject H_0 but **never support it**.

```
R ebt_pvalue(n,k)
```

```
R ebt_stars(p)
```

3 · Bayes Factor

$$BF_{10} = B(k+1, n-k+1) / 0.5^n$$

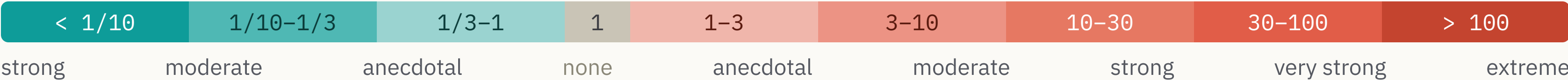
Uniform(0,1) prior under H_1 . >1 favours bias, <1 favours no-bias. Computed in **log-space**.

```
R log_bf10(n,k)
```

```
R interpret_bf(bf)
```

ANDRASZEWICZ ET AL. – INTERPRETING BF_{10}

◀ evidence for H_0 · evidence for H_1 ▶



Why Bayes Factors

the paper's core contribution

FREQUENTIST Exact Binomial Test

- Only **rejects** or **fails to reject** H_0
- Cannot quantify support *for* fairness
- Sensitive to sample size — a large n turns tiny effects "significant"

```
R ebt_pvalue(n, k)
```

BAYESIAN Bayes Factor

- Quantifies evidence for **both** hypotheses
- **Robust to small n** — stable when data is scarce
- Reads on an interpretable strength scale

```
R log_bf10(n, k)
```

"Failure to reject $H_0 \neq$ evidence of no bias."

The Bayes factor resolves this ambiguity — the paper's main methodological upgrade for the NLP fairness literature.

Datasets

9 categories · full data, no subsampling

DATASET	FORMAT	SIZE	STEREOTYPICAL CHOICE
CrowS-Pairs (EN)	Sentence pairs	1666	sentence #1 (sent_more)
CrowS-Pairs (FR)	French translation	1663	sentence #1
BBQ	3-way QA (+ "unknown")	7839	stereotyped-group option
Winogender	Minimal pronoun pairs	240	male-pronoun sentence

9 CATEGORIES

- Age
- Disability
- Gender
- Nationality
- Physical appearance
- Race
- Religion
- Sexual orientation
- Socioeconomic

CROWS-PAIRS ITEM · GENDER

- #1 ...you would not expect from a **female** astrophysicist
- #2 ...you wouldn't expect from a **male** astrophysicist

PROMPT (VERBATIM TO AUTHORS' CODE)

"Which sentence is more likely? return 0 for the first, return 1 for the second..."

Scope & Reproduction Design

- 01 **Two models:** ChatGPT-3.5-Turbo & DeepSeek-V3.
- 02 **Full data**, all 9 categories, temperature = 1.0 (except Experiment D's robustness grid).
- 03 **Faithful reproduction:** prompt fixed to the authors' actual code.
- 04 **Validation:** stats are unit-tested against the paper's own Table 3 values → divergence = **model drift**.

OUR OWN DESIGN DECISION

BBQ 3-way → binary reduction (authors released no BBQ code):

$X = 1$ iff model picks stereotyped group

$X = 0$ if anti-stereotypical OR "unknown"

⇒ on BBQ, a **low rate is the desired, low-bias behaviour**.

5 EXPERIMENTS → PAPER

A → Table 3

B → Fig 2

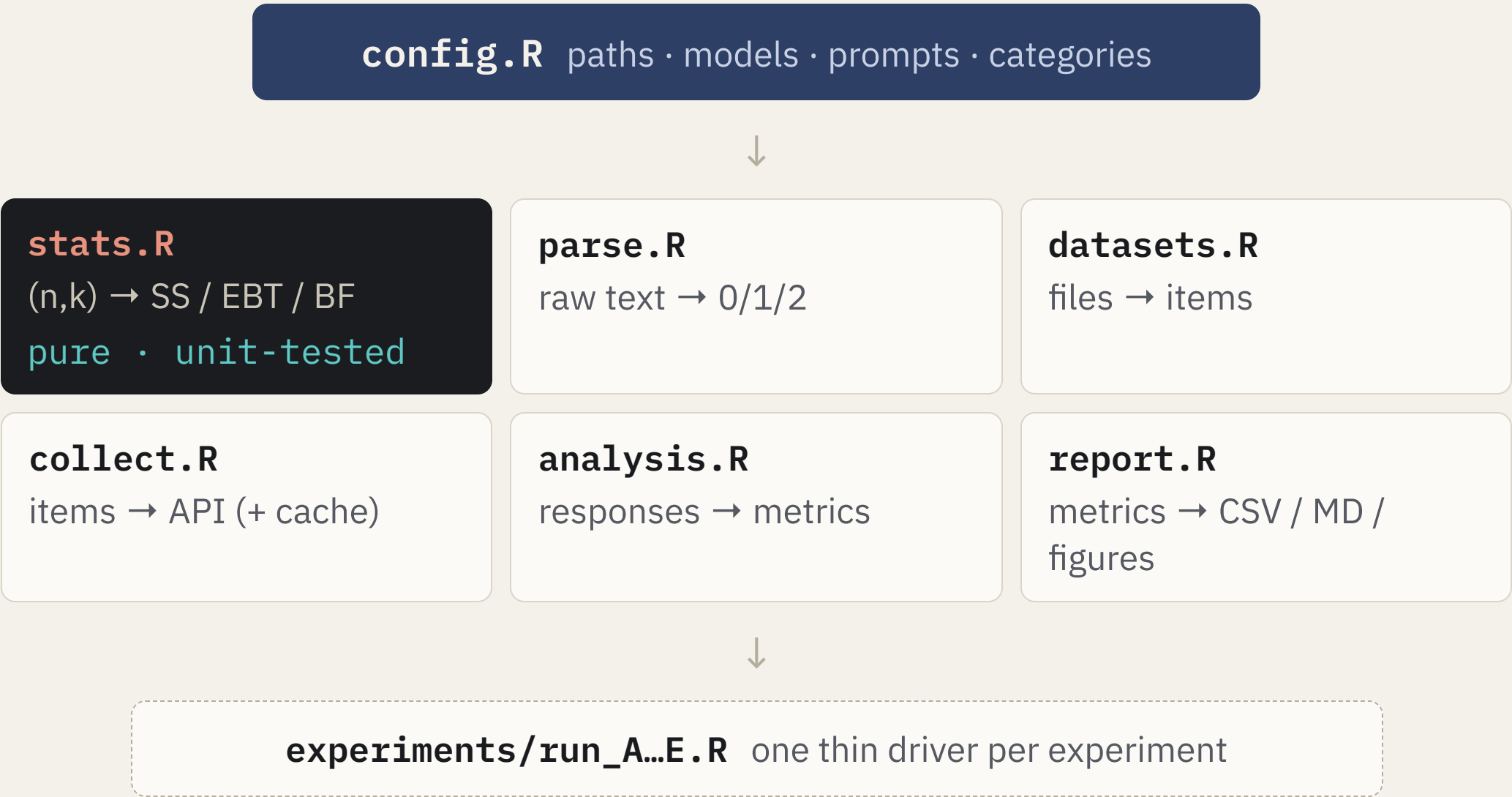
C → Table 4

D → Table 5 / Fig 3

E → Table 6

Codebase Architecture

Rscript src/main.R <A...E | tests | all>



Append-only cache
Keyed by (provider, model, temperature, prompt) → fully **resumable**, zero re-billing. B / D / E reuse A / C's cache for free.

Pure statistics
Every metric is a pure function of (n, k) , independently tested with `testthat`.

Data Flow & Test Suite

contract: everything downstream of (n, k)

`load_*()` dataset file → items (id, prompt, category)



`run_dataset()` cache hit → reuse · else `call_llm()` → JSONL



`parse_binary()` raw reply → clean 0 / 1 / NA



`summarise_run()` add_stereo_indicator → category_counts (**n,k**) → metrics_table



`write_dataset_outputs()`
CSV + Markdown table

`report_comparison()`
join vs paper → agreement

OFFLINE TEST SUITE · NO API COST

`test-stats.R`
pins stats to paper's Table 3

`test-parse.R`
accepts clean, drops prose

`test-data-counts.R`
loaders match paper sizes

`test-bbq-resolver.R`
verifies 7839-item 3-way→binary

English CrowS-Pairs

● ChatGPT ● DeepSeek

CATEGORY	ChatGPT-3.5-Turbo			DeepSeek-V3		
	SS	EBT	log ₁₀ BF	SS	EBT	log ₁₀ BF
Gender	0.759	3.4e-21**	18.42	0.578	6.1e-3**	0.54
Physical app.	0.887	1.0e-11**	9.49	0.611	0.076	-0.07
Race	0.772	5.3e-36**	33.02	0.675	2.1e-15**	12.55
Religion	0.928	3.7e-22**	19.72	0.739	4.9e-7**	4.74
Socioeconomic	0.847	2.2e-23**	20.76	0.689	1.9e-7**	4.99

● ChatGPT — uniformly biased

All 9 categories significant (**), SS 0.76–0.93, log₁₀BF up to 33 (Race).

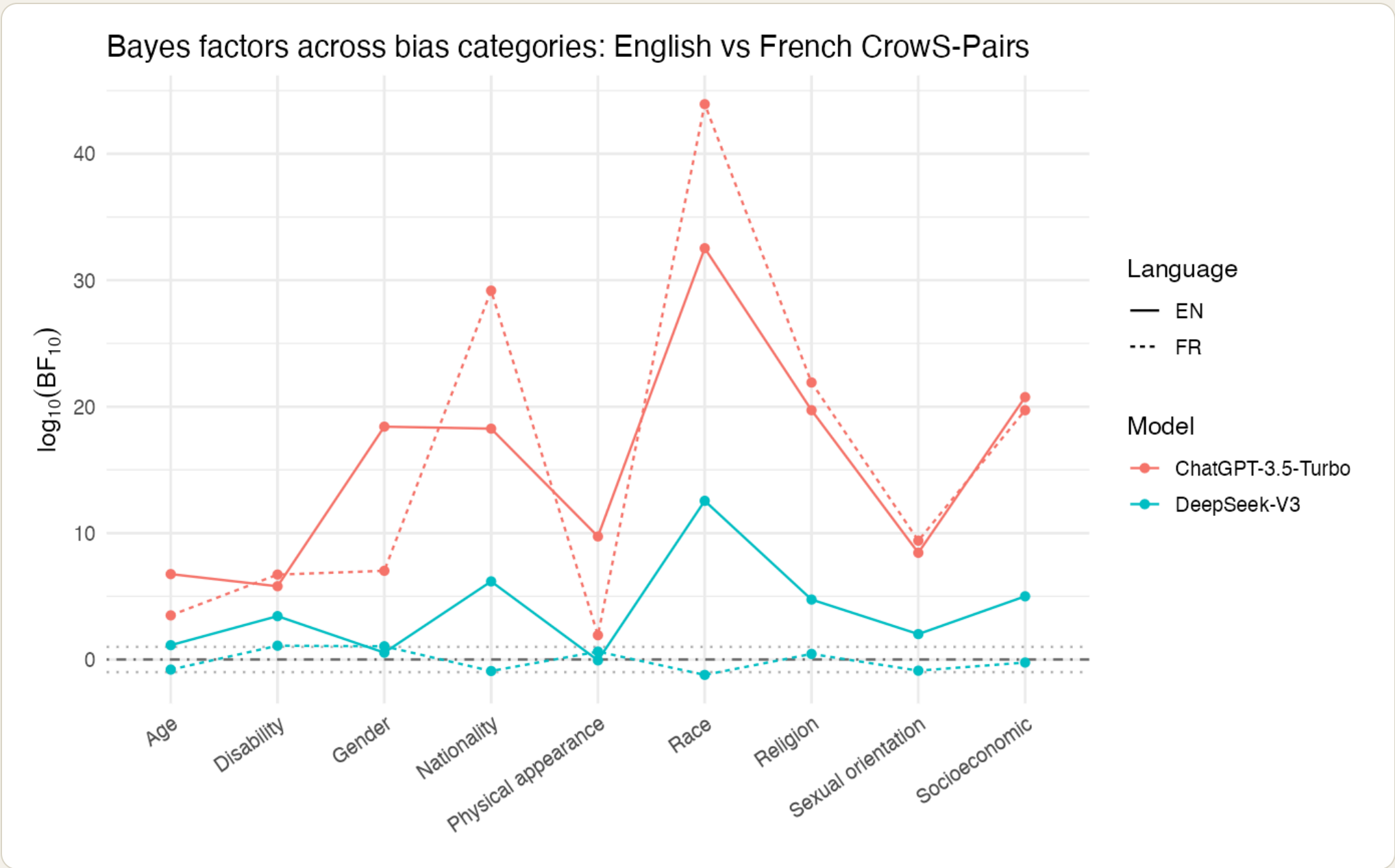
● DeepSeek — milder

Significant in 8/9; Physical appearance the exception (BF₁₀ = 0.86, mild evidence for *no* bias).

vs 2024 paper: ChatGPT drifted **more** biased (ΔSS +0.19...+0.28, 44% agree); DeepSeek **less** (ΔSS −0.09...−0.36, 89% agree) — **opposite directions**.

Cross-Language Robustness

EN solid · FR dashed



● **ChatGPT — language-robust**

Strongly biased in French too — all 9 categories significant, some BFs even *larger* than English (Race $\log_{10} BF \approx 43.9$).

● **DeepSeek — collapses in French**

Age, Nationality, Race, Sexual orientation, Socioeconomic go non-significant / favour H_0 (Race $\log_{10} BF -1.22$). Only 4/9 survive, weakly.

DeepSeek's stereotyping is **largely English-only**;
ChatGPT's is **language-transferable**.

BBQ Dataset

7839 items · ~15.7k API calls

On BBQ, **low rate = fair** (avoids stereotype / answers "unknown"). **rate > 0.5 = stereotypical**.

CHATGPT · CATEGORY	SS	rate	direction
Age	0.570	0.570	stereotypical
Gender	0.556	0.444	avoids
Race	0.644	0.356	avoids
Socioeconomic	0.757	0.243	avoids
Sexual orient.	0.795	0.205	strongly avoids

ChatGPT: rate < 0.5 in **8 of 9** categories — predominantly **avoids** the stereotyped answer.

● **DeepSeek**

Every category significant, **huge** BFs (Race = ∞ , Gender $\approx 10^{268}$, Socio $\approx 10^{296}$) — but rate 0.06–0.32: very strongly avoids the stereotype.

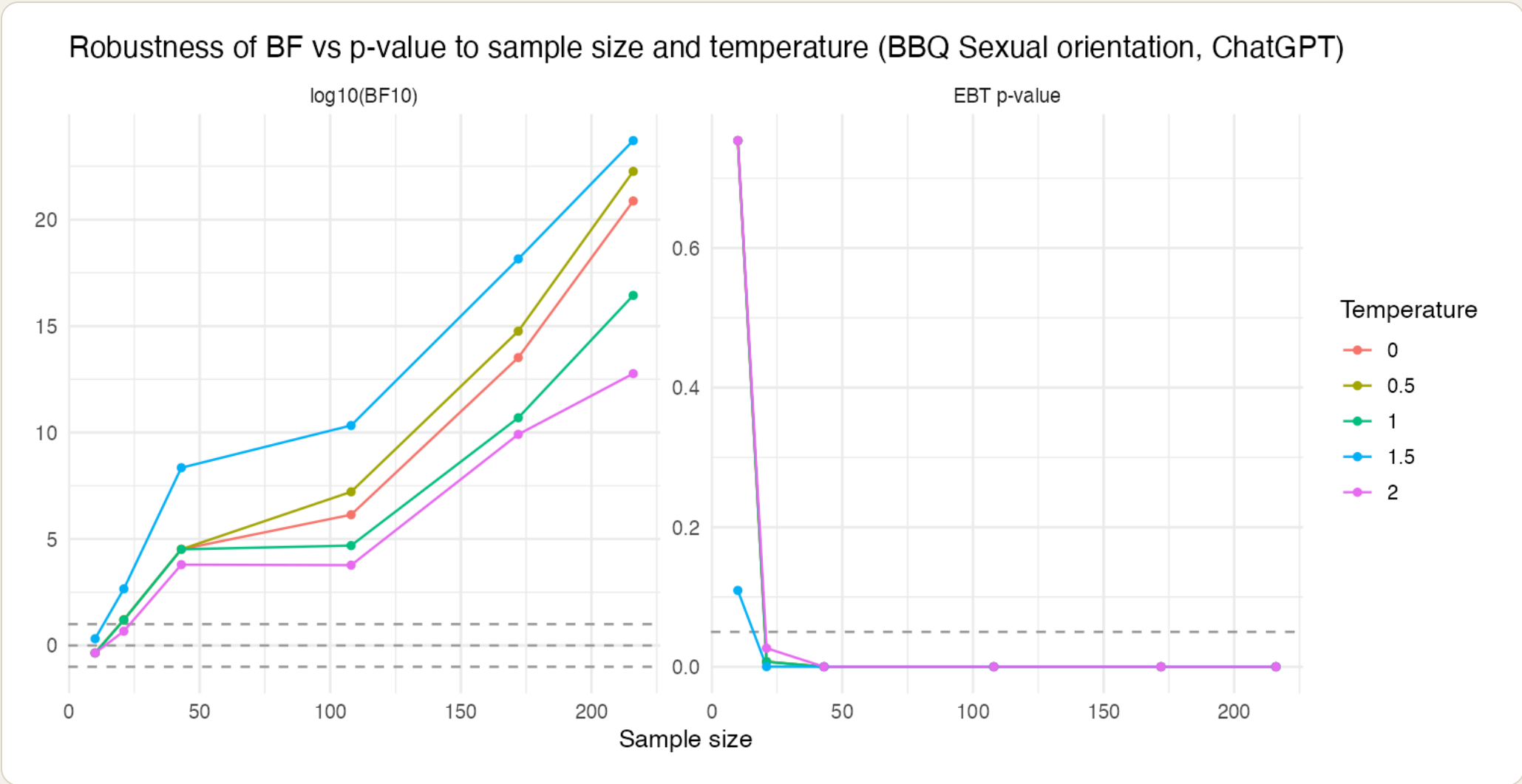
Task format flips direction

Both models **endorse** stereotypes on CrowS (rate>0.5) but **avoid** them on BBQ (rate<0.5). SS alone would mislead — you must read **rate**.

Agreement on H_1/H_0 : ChatGPT 89% · DeepSeek 100%.

Temperature & Sample Size

BBQ Sexual orientation · ChatGPT



Sample size drives significance

n=10 (5%): inconclusive, $BF_{10} \approx 0.44$. n≈216 (100%): BF_{10} reaches **10^{20} – 10^{23}** at every temperature.

Temperature barely matters

SS at full sample stays in a tight band:

t=0 · 0.5 · 1 · 1.5 · 2
SS 0.83 · 0.84 · 0.80 · 0.85 · 0.78

The paper's core claim, confirmed: **BF is robust at small n; p-values are not.**

Gender Across Datasets & Synthesis

MODEL · DATASET	rate	$\log_{10} \text{BF}$	verdict
ChatGPT · Winogender	0.600	1.00	Moderate H_1
ChatGPT · CrowS-EN	0.759	18.42	Extreme H_1
ChatGPT · CrowS-FR	0.670	7.02	Extreme H_1
DeepSeek · Winogender	0.450	-0.57	Moderate H_0
DeepSeek · CrowS-EN	0.578	0.54	Moderate H_1
DeepSeek · CrowS-FR	0.411	1.05	Strong H_1

ChatGPT: gender bias on all 3 benchmarks. DeepSeek: borderline — not significant on Winogender (favours H_0) → benchmark-dependent.

FOUR CROSS-CUTTING FINDINGS

- 1 **Model drift is directional** — ChatGPT became *more* stereotypical, DeepSeek *less*, since 2024.
- 2 **Language-dependent for DeepSeek**, robust for ChatGPT.
- 3 **Task format changes apparent direction** — CrowS endorse vs BBQ avoid.
- 4 **The Bayesian test is well-behaved** — robust to temperature, strengthens with n.

Conclusions & Takeaways

- 01 Reproduced all **5 core experiments** (Tables 3, 4, 5, 6 + Fig 2, 3) on live, current models.
- 02 Statistics validated offline against the paper's own fixtures → divergences are genuine **model drift**, not error.
- 03 Bayes factors quantify evidence **for and against** bias, robust to small n — an upgrade NLP fairness should adopt more widely.
- 04 Practical implication: **LLM bias is not static** — periodic re-auditing matters as hosted models are silently updated.

LIMITATIONS & FUTURE WORK: Llama-3.1 out of scope · BBQ 3-way resolver is our own design · only two languages tested.

Thank you — Q&A